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SIRT1 Exon 2-Encoded IDR Gates CREB-Dependent Transcriptional Timing During Fasting

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ABSTRACT

Metabolic adaptation to fasting requires precise temporal control of gene expression. Although transcription factors and upstream co-regulators that mediate the fasting response are well studied, the role of intrinsically disordered regions in coordinating transcriptional timing remains unclear. Here, we identify the exon 2-encoded intrinsically disordered region of SIRT1 as a regulator of fasting-responsive transcription in the liver. Using a physiological SIRT1$^{\Delta E2}$ mouse model, we show that loss of this region alters the sequence and magnitude of starvation-induced transcriptional responses, leading to premature activation of gluconeogenic genes and excessive hepatic glucose output. Mechanistically, exon 2 loss weakens SIRT1 interaction with CREB and reduces the ability of SIRT1 to restrain CREB-dependent transcription, while also affecting interactions with FOXO1 and PPARα. These changes impair fasting adaptation and glucose homeostasis. Together, our findings identify the SIRT1 exon 2-encoded IDR as a non-catalytic regulatory element that contributes to the timing of hepatic transcriptional responses during nutrient stress.

1 | Introduction

Precise spatial and temporal regulation of transcription is essential for adapting gene expression during physiological transitions and environmental changes. In mammals, transcriptional complexity arises from cis-regulatory elements with multiple TF binding sites and upstream co-regulators that modulate TF activity [1–3]. While much is known about TF–DNA and TF–co-regulator interactions, emerging work highlights a critical role for IDRs within TFs that influence chromatin scanning, promoter selectivity, and target gene activation [4–6].

IDRs have redefined the mechanistic underpinnings of cis-trans and trans-trans interactions, crucial for physicochemical organization of transcription, including spatial control via phase separation and condensates. Independently, while promoter specificity is dictated by DNA-binding domain (DBD), IDRs have been established as key determinants of promoter selectivity or promiscuity [7–10]. Theoretical/Heuristic approaches, besides others, have clearly provided unique insights in this context. Importantly, thermodynamic frustration is known to contribute to transcriptional output by enabling TF interactions with multiple partners (Mediators, co-activators, co-repressors, etc.) to exert context-dependent regulation [11–14]. However, most of our current understanding is limited to the role of IDRs in TFs with a paucity of information vis-à-vis regulation stemming from disorder-dependent dynamic interactions in upstream factors such as p300/CBP, SIRT1, and others. To reiterate, even though factors like KATs (TIP60, GCN5, p300, etc.) and KDACs (Sirtuins, HDAC1, HDAC3, etc.) are pivotal for transcription factor activity, synergistic or antagonistic interactions between them that are possibly mediated by IDRs in KATs/KDACs are less understood.

Despite this there are two key aspects which seem to have been largely unanswered in the field. Firstly, it is interesting to note that eukaryotic genomes have complex *cis*-regulatory landscapes (enhancers, promoters, insulators) and IDRs have been speculated to have co-evolved with these elements to manage increasing regulatory demands [15–18]. But causal evidence to demonstrate this co-evolution in transcriptional machinery is non-trivial. Second, and probably more important, aspect is the bi-directional interplay between IDRs and physiology. This is key because most biological processes involve physiological transitions that entail transcriptional rewiring. Even though IDRs have been recognized as drivers of dynamic protein–protein interactions, necessary for transcription, if/how these influence or control temporal cascades of gene expression remains poorly understood. Given this context, we hypothesize that IDRs in upstream regulatory factors act as conformational timers to not only orchestrate temporal gene expression but also to buffer aberrant cis-trans interactions on promoters with multiple TF binding sites.

Given this context, starvation offers an ideal model to investigate how intrinsically disordered regions (IDRs) in upstream regulators influence transcriptional timing in vivo. Physiological transitions such as development, immune responses, and metabolic adaptation require dynamic gene regulation by multiple transcription factors (TFs) acting in sequence or combination. Although overlapping regulatory inputs are well known, how TF modules are temporally ordered or exchanged remains unclear. The liver's fasting response, particularly gluconeogenesis, involves TFs such as CREB, FOXO1, HNF4α, and the glucocorticoid receptor [19–22]. While each is essential, their sequential activation during fasting is poorly understood. SIRT1, a NAD^+-dependent deacetylase, plays a key role in this process. Unlike other metabolic sensors, SIRT1 contains extended N- and C-terminal IDRs. Its N-terminal IDR, encoded by exon-2, interacts with starvation-responsive TFs, but its in vivo role in coordinating transcriptional transitions is not fully known [23–27]. Understanding this may reveal broader principles of dynamic transcriptional control during nutrient stress.

In this study, we examine the role of the SIRT1 N-terminal IDR, encoded by exon-2, in regulating transcriptional dynamics during fasting. Using a conditional $SIRT1^{\Delta E2}$ mouse model, we dissect its in vivo contribution to hepatic gene regulation and systemic glucose homeostasis. We find that the IDR temporally gates SIRT1 interactions with key starvation-responsive TFs, notably CREB. Loss of this region results in premature and amplified activation of gluconeogenic genes, increased pCREB occupancy on chromatin, and excessive hepatic glucose output, culminating in impaired fasting-refeeding transitions. These findings identify the SIRT1 IDR as a molecular timer that ensures transcriptional responses are appropriately phased. More broadly, this work highlights the regulatory potential of IDRs in upstream metabolic sensors, expanding their functional relevance beyond TFs to include fine-tuning of transcriptional networks during physiological transitions.

2 | Materials and Methods

2.1 | Animal Experiments

All the experiments and protocols were done in accordance with the Institutional Animal Ethics Committee (IAEC) guidelines and approvals. Mice were housed at standard housing conditions of 12 h light/dark cycles (at TIFR Mumbai, CCMB, IISER Pune animal houses). The temperature was maintained at 20°C ± 2°C and humidity at 50 ± 20. They were fed a normal chow diet and had free access to autoclaved drinking water, unless otherwise mentioned. Mice strains include C57BL/6NCrl, $SIRT1^{E2.LOX/LOX}$ (generated as described), $ERT2^{CRE/CRE}$ (Jax Stock ID: 008463), $Alb^{CRE/CRE}$ (Jax Stock ID: 018961), $SIRT1^{E2.LOX/LOX}$; $ERT2^{CRE/CRE}$, $SIRT1^{E2LOX/LOX}$; $Alb^{CRE/CRE}$. SIRT1-Lox-E2-Lox mouse was generated by Turku Center for Disease Modeling University of Turku, Turku, Finland as a paid service. All experimental animals were euthanized by cervical dislocation and the relevant tissues were isolated and snap frozen in liquid nitrogen, unless stated otherwise. Littermates of the same genotype and sex were randomly assigned to the experimental groups.

2.2 | Tamoxifen Administration for Generating Conditional Whole Body SIRT1 Exon-2 Knockout Model ($SIRT1^{\Delta E2}$)

To obtain whole-body conditional knockout of Exon-2 of SIRT1, 2-month-old male mice which were homozygous for E2 LOX and ERT2 Cre with the genotype $SIRT1^{E2.LOX/LOX}$; $ERT2^{CRE/CRE}$ were administered Tamoxifen by oral gavage. Tamoxifen was reconstituted in corn oil at a concentration of 20 mg/mL by dissolving at 37°C, and 4 doses of 100 mg/kg body weight were administered on alternate days. Knockout was confirmed from tail clip DNA after 7 days from the last dose. Animals were used for experiments as per the paradigms used and explained below and in relevant sections.

2.3 | Generation of SIRT1 Exon-2 Liver Knockout

To obtain Liver-specific knockout of exon-2 of SIRT1, the $SIRT1^{E2.LOX/LOX}$ were crossed with albumin Cre mice where the CRE is under the albumin promoter and hence the KO is restricted to the liver. The mice line was maintained as heterozygous $SIRT1^{E2LOX/+}$; $ALB^{cre/+}$ and with the help of tail clip genotyping the $SIRT1^{hep\text{-}\Delta E2}$ (LKO) and $SIRT1^{E2.LOX/LOX}$ (WT) animals were selected for the experiments described later.

2.3.1 | Glucose Tolerance Test

For glucose tolerance test (GTT), mice were starved for 12 h and body weight were measured, baseline blood glucose was measured from tail prick using a glucometer (AccuCheck active) labeled as zero minutes. 2 g/kg body weight of glucose was injected intraperitoneally (I.P.). Subsequently, the blood glucose was monitored at 15, 30, 60, and 120 min, post the glucose I.P. injection. The data was plotted using GraphPad-Prism 8.0.

2.3.2 | Pyruvate Tolerance Test

For the pyruvate tolerance test (PTT), mice were starved for 6, 12 or 24 h (depending on the experimental requirement) and body weight was measured, baseline blood glucose was measured from tail prick using a glucometer (AccuCheck active) labeled as zero minutes. 2 g/kg body weight of pyruvate was injected intraperitoneally (I.P.). Subsequently, the blood glucose was monitored at 15, 30, 60, and 120 min post the pyruvate I.P. injection. The data was plotted using GraphPad-Prism 8.0.

2.3.3 | Primary Hepatocyte Culture

Male SIRT1$^{\Delta E2}$ and SIRT1$^{E2LOX/LOX}$ mice (12–16 weeks old) were anesthetized by intraperitoneal injection of Thiopentone (Neon Laboratories Ltd., Mumbai, India). Once sedation was confirmed, the mice were disinfected with 70% ethanol. Liver perfusion was performed through the inferior vena cava/portal vein using 30 mL of Hank's Balanced Salt Solution (HBSS) (5.33 mM potassium chloride, 0.44 mM KH_2PO_4, 4.16 mM $NaHCO_3$, 137.93 mM NaCl, and 0.338 mM Na_2HPO_4, pH adjusted to 7.4). The solution was supplemented with 5.5 mM glucose (Sigma-Aldrich G8769), 25 mM HEPES (USB 16926), and 100 mM EGTA (Sigma-Aldrich E3889). Following perfusion, the liver was digested using a digestion medium containing DMEM-Low Glucose (Sigma-Aldrich D5523), 15 mM HEPES, antibiotic-antimycotic (Anti-Anti), and collagenase type IV (Sigma-Aldrich C5138). The digested liver tissue was carefully minced and filtered through a 70 μm cell strainer to obtain a hepatocyte suspension. The cell suspension was then centrifuged at rcf value of 50 g for 5 min at 4°C. The resulting pellet was washed thrice with high-glucose DMEM (25 mM glucose; Sigma-Aldrich D7777) supplemented with 10% fetal bovine serum (Gibco 16000044). The cells were counted using a hemocytometer and plated onto collagen-coated (Sigma-Aldrich C3867) 100 mm/6 well culture dishes at a density of 2×10^6 or 3×10^5 cells per plate, respectively. The cells were incubated at 37°C in a humidified atmosphere with 5% CO_2 to allow for adherence. Six hours post-plating, the medium was replaced to remove non-adherent cells and replaced with high-glucose DMEM without any serum addition.

2.3.4 | Glucagon Treatments in Hepatocytes

Primary hepatocytes were isolated as described above, and the cell medium was changed to DMEM high glucose without serum for 12 h. To expose hepatocytes to starvation-like conditions, the medium was changed to DMEM-No glucose media (supplemented with glutamine 2 mM and pyruvate 2 mM) with the final concentration of glucagon at 30 nM for the time points as mentioned. Post-treatment, the cells were collected in RIPA buffer or TRIzol as per the experiment's requirement. The hepatocytes were metabolically synchronized for pulsatile glucagon experiments by replenishing the medium with HG-DMEM 3 h before media change, followed by media change into DMEM-NO Glucose (±glucagon). Every media change was preceded by two phosphate buffer saline washes (PBS) (Thermo, #88512) to remove the residual media.

2.4 | CREB Inhibitor (666-15) and PKA Inhibitor (H89) Treatments for Hepatocytes

Primary hepatocytes were isolated as described above, and the cell medium was changed to DMEM high glucose without serum for 12 h. To assess the effect of CREB and PKA inhibitor, the hepatocytes were metabolically synchronized by replenishing the medium with HG-DMEM 3 h before media change, followed by media change into the following: HG, DMEM-NO Glucose + glucagon (30 nM), DMEM-NO Glucose + glucagon (30 nM) + 666-15 (1 μM), and DMEM-NO Glucose + glucagon (30 nM) + H89 (10 μM). Cells were collected after 3 h of treatment. Every media change was preceded by two phosphate buffer saline washes (PBS) (Thermo, #88512) to remove the residual media. Post-treatment, the cells were collected in RIPA buffer or TRIzol as per the experiment's requirement.

2.4.1 | Plasmid Transfection and Luciferase Assay in Hepatocytes

Primary hepatocytes were isolated from Male SIRT1$^{\Delta E2}$ and SIRT1$^{E2LOX/LOX}$ mice and plated as described above. Plasmid transfections were done using the JetPrime transfection reagent (Polyplus, #101000027) for 24–48 h as per the manufacturer's protocol. After 6 h of plating, hepatocytes were transfected with a plasmid featuring the CREB promoter upstream of the luciferase gene, a kind gift from Prof. Mark Montminy, along with a β-galactosidase expressing plasmid. After 36 h of transfection, the cells were treated with DMEM-NG + 30 nM glucagon for 3 h. The hepatocytes were washed with chilled PBS and harvested in passive lysis buffer (Promega # E194A) for 15 min, followed by centrifugation at 12 000 rpm. The supernatant was collected for the BCA estimation, β-Gal assay and the luciferase assay. β-Gal assay was performed using the ONPG (Sigma #N1127) method, where 30 μL of supernatant is incubated with 50 μL of ONPG (4 mg/mL) at 37°C for 20–30 min. 100 μL of stop solution (1 M Na_2CO_3) was added to wells, and the absorbance was measured at 420 nm. For the luciferase assay, 30 μL of supernatant was taken in a black plate to this 100 μL of luciferase assay system reagent (Promega #E1501) was added. The counts for the luminescence were measured in the TriStar LB941 (Berthold technologies) and normalized to the β-Gal activity and represented as fold change.

2.4.2 | Mass Spec Sample Preparation

2.4.2.1 | Polar Non-Derivatized Metabolite Extraction. A known weight of liver tissue was homogenized in 75% ethanol containing internal standards: C13-Glucose (10 nM; Cambridge Isotopes, #CLM-1396) and L-Tryptophan (1.5 nM; Cambridge Isotopes, #DLM-615862). The samples were vortexed for 3 min, incubated at 80°C for 3 min, and subsequently transferred to ice for 5 min. Following incubation on ice, the samples were centrifuged at 17 000 g for 10 min at 4°C.

An equal volume of the resulting supernatant from all samples was transferred to fresh Eppendorf tubes and dried using a speed vacuum concentrator. The dried samples were stored at −20°C until mass spectrometry analysis was performed.

2.4.2.2 | Polar Derivatized (TCA) Metabolites Extraction. A measured quantity of liver tissue was homogenized in 75% ethanol containing internal standards: D4-succinic acid (3 nM; Cambridge Isotopes #DLM-2307). The mixture was vortexed for 3 min, incubated at 80°C for 3 min, and then incubated on ice for 5 min. After cooling, the samples were centrifuged at 17000 g for 10 min at 4°C. The supernatant was collected in equal volumes into fresh Eppendorf tubes and dried using a speed vacuum concentrator.

The dried samples were re-suspended in a 1:2 ratio of H_2O and 1 M N(3-dimethylaminopropyl)-N′-ethylcarbodiimide (prepared in 13.5 mM pyridine buffer, pH 5.0). The samples were mixed by tapping, followed by the addition of 150 μL of 0.5 M O-benzyl hydroxylamine (prepared in 13.5 mM pyridine buffer, pH 5.0). The samples were incubated at 25°C for 1 h.

After incubation, 350 μL of ethyl acetate was added, and the mixture was shaken for 10 min. The samples were centrifuged at 3000 g for 5 min at 4°C, and the supernatant was transferred to a fresh tube. This ethyl acetate extraction process was repeated two more times. The collected supernatant was then dried using a speed vacuum concentrator. The final dried samples were stored at −20°C until analysis by mass spectrometry.

2.4.3 | Mass Spectrometry Analysis

Dried samples were analyzed using an Agilent 6546 Quadrupole Time-of-Flight (QTOF) mass spectrometer operating in Auto-MS/MS acquisition mode, coupled with an Agilent 1290 Infinity II UHPLC system. Liquid chromatography (LC) separation was performed using a Synergi 4 μm Fusion-RP 80 Å LC column (100 × 4.6 mm, Ea; catalogue no: 00D-4424-E) with a Gemini guard column (Phenomenex, 4 mm × 3 mm; catalogue no: KJ0-4282).

For polar non-derivatized metabolites, chromatographic separation was conducted at a flow rate of 0.5 mL/min. The mobile phase composition for positive ion mode consisted of solvent A: 99.9% (v/v) H_2O + 0.1% formic acid, and solvent B: 99.9% (v/v) methanol (MeOH) + 0.1% (v/v) formic acid. For negative ion mode, solvent A was 100% (v/v) H_2O with 5 mM ammonium acetate, while solvent B was 100% (v/v) acetonitrile (ACN). The gradient elution program was as follows: 0–3 min, 5% solvent B; 3–10 min, 5%–60% solvent B; 10–11 min, 60%–95% solvent B; 11–14 min, 95% solvent B (wash step); 14–15 min, 95%–5% solvent B; and 15–21 min, 5% solvent B (re-equilibration step).

For polar derivatized metabolites, data acquisition was performed in positive ion mode with a flow rate of 0.5 mL/min. The mobile phase consisted of solvent A: 99.9% (v/v) H_2O + 0.1% formic acid, and solvent B: 99.9% (v/v) MeOH + 0.1% (v/v) formic acid. The LC run time was 30 min, with the following gradient elution program: 0–3 min, 0% solvent B; 3–10 min, 0%–5% solvent B; 10–11 min, 5%–60% solvent B; 11–18 min, 60%–95% solvent B; 18–19 min, 95% solvent B (wash step); 19–22 min, 95%–60% solvent B (re-equilibration step); 22–25 min, 60%–5% solvent B; and 25–30 min, 5%–0% solvent B.

The autosampler and column temperatures were maintained at 10°C and 40°C, respectively. Mass spectrometric data were acquired using a dual Agilent Jet Stream electrospray ionization (AJS ESI) source in both positive and negative ion modes. The MS operating parameters were as follows: drying gas temperature, 320°C; sheath gas temperature, 320°C; drying gas flow, 11 L/min; sheath gas flow, 11 L/min; nebulizer pressure, 45 psi; capillary voltage, 3 kV; nozzle voltage, 1 kV; and fragmentor voltage, 150 V. Precursor ion selection was performed with a maximum of 10 precursor ions per cycle, and collision energy was set at 5 eV.

2.5 | Data Analysis and Quantitation

LC–MS data processing was performed using Agilent MassHunter Qualitative Analysis 10.0. Identified peaks were manually verified based on retention times and fragment ions obtained from MS/MS spectral analysis. A mass accuracy threshold of 10 ppm was applied to all analytes across all experiments described in this study. Metabolite quantitation was conducted by calculating the area under the curve (AUC) of each analyte relative to its corresponding internal standard. The resulting values were subsequently normalized to the total protein content within liver tissue samples to ensure accurate absolute concentration measurements.

2.5.1 | RNA Isolation and Quantitative PCR (qPCR)

Total RNA was isolated from tissue or cells using TRIzol following the manufacturer's instructions. Briefly, TRIzol was added to cells or tissues and homogenized using a pestle. Following 5 min of incubation at room temperature, 0.2 mL of chloroform was added per 1 mL of TRIzol. After shaking, samples were incubated for 2 min and centrifuged at 12000 rpm for 15 min at 4°C. The aqueous phase was collected and transferred into a new microcentrifuge tube, and isopropanol was added in the ratio of 1–2 volumes of TRIzol, followed by 3–4 h of incubation at −20°C. Samples were centrifuged at 12000 rpm for 15 min and RNA pellets were washed using 70% ethanol. After washing, the RNA pellets were dried at 37°C for 3–5 min and resuspended in the desired volume of DEPC water. The concentration and purity of RNA samples were measured using UV spectroscopy (by measuring the absorbance of samples at 260 and 280 nm).

1–5 μg RNA was utilized for cDNA preparation using Random Hexamer and SuperScript IV Reverse Transcriptase following the manufacturer's protocol. Quantitative PCR was performed using the LightCycler 96 (Roche) and the KAPA SYBR FAST qPCR Master Mix (2X) Kit following the manufacturer's guidelines. The expression of target genes was normalized with actin transcript levels.

2.5.2 | Lysate Preparation

Cells or tissue were homogenized in ice-cold RIPA buffer (50 mM Tris pH 8.0, 150 mM NaCl, 0.1% SDS, 0.5% Sodium deoxycholate, 1% Triton X-100, 0.1% SDS) supplemented with 1 mM PMSF, 1.5XProtease inhibitor cocktail, 1.5XPhostop and incubated for 10–20 min with brief vortexing, occasionally. The lysates were sonicated using Bioruptor Pico (Diagenode) (5 cycles for cells and 10 cycles for tissues with 30 s ON/30 s OFF at 4°C) followed by centrifugation at 13 000 rpm for 15 min at 4°C. Supernatants were transferred into fresh microcentrifuge tubes and used for estimating protein concentration using BCA (Bicinchoninic Acid) Protein estimation assay as per the manufacturer's instruction. Samples were boiled at 95°C for 10 min after adding Laemmli sample buffer (50 mM Tris–Cl pH 6.8, 2% SDS, 0.1% bromophenol blue, 10% glycerol, 100 mM DTT).

2.5.3 | Western Blotting

40–60 μg of equal protein were run on sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and transferred to polyvinylidene fluoride (PVDF) membrane. After blocking with 5% milk in Tris-buffered saline with 0.1% Tween (TBST) for 90 min, desired proteins were probed by overnight incubation of the membrane with primary antibodies (diluted in 5% BSA in TBST) at 4°C with continuous rocking. Following three washes with TBST at room temperature, the membranes were incubated with horseradish peroxidase-conjugated secondary antibodies for one and a half hours at room temperature. Membranes were washed thrice with TBST at room temperature, and the bands of proteins were visualized in the GE Amersham Imager 600 instrument (General Electric) using a chemiluminescent-based detection kit. The band intensities of targeted proteins were measured using ImageJ and normalized with that of the loading control. Expression of actin or tubulin, or histone H3 were utilized as a loading control, as mentioned in the respective figures.

2.5.4 | Immunoprecipitation (IP)

50 μg of liver tissue from the mice as per the experiment or plated hepatocytes as per the treatment were collected in PBS, washed twice and lysed in TNN buffer (50 mM Tris–cl pH 7.5, 150 mM Nacl, 0.9% NP40), along with 1.5× PIC, 1.5× Phosstop and 1.5 mM PMSF and incubated for 15 min. The lysate was centrifuged at 12 000 rpm at 4°C. Supernatant was carefully separated and protein A beads were used for preclearing at 4°C at slow rotor (5 rpm) for 1.5 h. Following preclearing, input was separated and 3–4 mg of protein was used for IP. Control antibody (Rabbit IgG) and SIRT1 were used for IP. Immunoprecipitation was allowed for 8–10 h at 4°C on at slow rotor followed by pull down with Prot A magnetic beads. After 3–4 h, the unbound fraction was collected and the beads were proceeded for washing. Post the washes, the beads were magnetized, all the residual buffer removed, and laemmli buffer was added to the beads, vortexed for 5 s and boiled for 15 min. After boiling, the beads were magnetized again, and the supernatant was carefully separated. All the fractions were analyzed as per the experiments need and mentioned in the relevant results section.

2.5.5 | Chromatin Immunoprecipitation (ChIP) From Hepatocytes

2.5.5.1 | Crosslinking. Hepatocytes (2×10^6) were plated in 100 mm collagen-coated cell culture-treated Petri dishes. On the second day post-plating, both SIRT1$^{E2.Lox/Lox}$ and SIRT1$^{\Delta E2}$ cells were treated with starvation mimicking media (NG + glucagon 30 nm). After 3 h of treatment, cells were washed twice with ice-cold PBS and crosslinked with 1% formaldehyde for 8 min at room temperature. Crosslinking was quenched with 125 mM glycine for 5 min, followed by three washes with chilled PBS. Cells were scraped in cold PBS, collected by centrifugation at 1000 rpm for 5 min, the supernatant was discarded, and the resulting pellet was stored at −80°C for further processing.

2.5.5.2 | Lysis and Sonication. Frozen cell pellets were thawed and resuspended in 600 μL of ice-cold Cell Lysis Buffer (CLB; see Table S1). After incubation on ice for 10 min, samples were centrifuged at 3500 rpm for 5 min at 4°C, and the supernatant was discarded. The pellet was resuspended in 500 μL of CLB, vortexed briefly, and incubated on ice for another 5 min. This step was repeated once more with 300 μL of CLB. The final pellet was resuspended in Nuclear Lysis Buffer (NLB), incubated on ice for 10 min, and then diluted with 500 μL of ChIP Dilution Buffer. Samples were snap-frozen in liquid nitrogen and thawed on ice prior to sonication.

Sonication parameters were optimized to obtain chromatin fragments suitable for transcription factor (TF) ChIP. Post-sonication, samples were centrifuged at 13 000 rpm for 15 min at 4°C, and the supernatant (sheared chromatin) was collected for immunoprecipitation and quality assessment.

2.5.5.3 | Reverse Crosslinking (For Chromatin Assessment). To assess chromatin quality, 20 μL of the sheared chromatin was diluted with 180 μL of ChIP Dilution Buffer. To this, 8 μL of 5 M NaCl was added, and the mixture was incubated at 65°C for 12–20 h. RNase A (20 μL of 10 mg/mL) was added, vortexed, and incubated at 37°C for 2 h, followed by the addition of 5 μL of Proteinase K and incubation at 45°C for 2–4 h. DNA was purified using a phenol:chloroform extraction (1:1), followed by a chloroform wash and ethanol precipitation with 2 volumes of chilled ethanol and 25 μL of 3 M sodium acetate. Precipitated DNA was centrifuged at 13 000 rpm at 4°C for 30 min, washed with 70% ethanol, and air-dried. DNA was finally resuspended in 20 μL of nuclease-free water and assessed for concentration, purity, and fragment size before proceeding with ChIP.

2.5.5.4 | Bead Preparation. Protein A magnetic beads were washed with PBST (PBS + 0.1% Tween-20), then resuspended in ChIP Dilution Buffer supplemented with 10 μL of 10 mg/mL tRNA and 1% BSA. Beads were blocked by incubation at 4°C with gentle rotation for 4 h.

2.5.5.5 | Chromatin Immuno-Precipitation (ChIP). Approximately 35–50 μg of chromatin per immunoprecipitation (IP) was used. Blocked beads were added for pre-clearing at 4°C for 2 h with gentle rotation. After pre-clearing, 5% of the input was reserved, diluted to 200 μL with ChIP Dilution Buffer, and stored at −20°C. The remaining pre-cleared chromatin was split into two aliquots: one incubated with rabbit IgG (negative control) and the other with anti-pCREB antibody. Antibody binding was performed overnight (10–12 h) at 4°C with gentle rotation. Subsequently, blocked beads were added to each tube and incubated for an additional 4 h.

Beads were sequentially washed twice with the following buffers, each for 5 min at 4°C: Low Salt Buffer, High Salt Buffer, Lithium Chloride Buffer, and finally Tris-EDTA Buffer. Following washes, 100 μL of Elution Buffer was added, and the beads were vortexed and incubated at 65°C for 30 min. The supernatant was collected, and the elution was repeated with an additional 100 μL of Elution Buffer. Supernatants were pooled, and 8 μL of 5 M NaCl was added to both IP and input samples for reverse crosslinking, followed by overnight incubation at 65°C.

2.5.6 | DNA Purification and qPCR

Reverse crosslinking was followed by RNase A and Proteinase K digestion, as described previously. Glycogen blue and 25 μL of 3 M sodium acetate were added, and DNA was precipitated overnight at −20°C. After ethanol precipitation and drying, the DNA pellet was resuspended in 50 μL of nuclease-free water. The immunoprecipitated DNA was analyzed by qPCR using primers for target gene loci.

2.5.6.1 | Gluconeogenesis Assay. Hepatocytes (50 000 cells/well) isolated from SIRT1$^{E2.Lox/Lox}$ and SIRT1$^{\Delta E2}$ mice were seeded in collagen-coated 24-well plates. Following adherence, cells were treated according to the experimental conditions—either with normal glucose (NG) medium alone or NG supplemented with 30 nM glucagon—for 6 or 12 h. Prior to treatment, cells were thoroughly washed with PBS to remove residual serum or media components.

At the end of the treatment period, supernatants were collected for glucose measurement, and the corresponding wells were used to scrape cells for protein quantification by BCA. The glucose concentration in the culture supernatant reflects the extent of hepatocyte-mediated gluconeogenesis.

Glucose levels were quantified using the Glucose Colorimetric Detection Kit (EIAGLUC, Invitrogen) following the manufacturer's protocol. This assay is based on the enzymatic oxidation of glucose by glucose-oxidase, producing hydrogen peroxide, which in the presence of horseradish peroxidase (HRP) reacts with a chromogenic substrate to yield a colored product measurable at 560 nm.

For quantification, assay standards provided in the kit were serially diluted to prepare the following concentrations: 32, 16, 8, 4, and 2 mg/mL. In each well, 20 μL of either sample or standard was added in duplicate. To this, 25 μL of HRP solution was added, followed by 25 μL of chromogenic substrate, and finally 25 μL of 1× glucose oxidase solution. The reaction mixtures were incubated at room temperature for 30 min.

Absorbance was measured at 560 nm using a microplate reader. A standard curve was generated from the known glucose standards, and sample concentrations were calculated accordingly.

2.5.6.2 | Immunofluorescence in Hepatocytes. Hepatocytes (50 000 cells/well) isolated from SIRT1$^{E2.Lox/Lox}$ and SIRT1$^{\Delta E2}$ mice were seeded in collagen-coated coverslips. Following adherence, cells were treated according to the experimental conditions—either with High glucose (HG-25 mM glucose) medium or no glucose (NG) supplemented with 30 nM glucagon—for 3 h. After the treatment, cells were washed with chilled PBS and were then immediately fixed in 4% formaldehyde (methanol-free) (Thermo Scientific Cat. No # 28908) for 20 min at room temperature. Post fixation, cells were washed gently with 1× PBS thrice and blocked for 1 h in 5% BSA in TBST (0.05% Triton). Cells were then incubated with SIRT1 antibody (CST) at a concentration of 1:1000 overnight. Cells were washed thrice with 1× PBS and incubated with Alexa Fluor-conjugated secondary antibodies (Thermo-Fisher Scientific, AF 488 or AF 647) for 1 h at room temperature. Cells were washed three times with 1× PBS and stained with DAPI (1:1000 of 1 mg/mL). The coverslips were mounted in Vectashield on a glass slide and kept in place using transparent nail paint. They were taken for imaging post-drying in the FV3000 confocal system at 60× magnification with oil (Olympus).

2.6 | Simulation and Analysis Method

The initial configurations for Sirtuin1 were obtained from RoseTTAFold [28], as no structure for the domains of interest is available in the RCSB PDB database. We used the primary sequence of Sirtuin1 corresponding to UniProt IDs Q923E4 and Q3UNI1 to generate the initial configurations for the exon-1 (E1), exon-2 (E2), and exon-3 (E3) domains. The Molecular Dynamics (MD) simulations of the Sirtuin1-E1E2E3 (hereby termed as E1E2E3) and Sirtuin1-E1E3 (hereby termed as E1E3) were performed using the GROMACS package version 20xx [29–32] with the a99SB-disp force field and a99SB-disp water model [33]. The system was neutralized using NaCl and the salt concentration was kept at 150 mM for all our simulations. First, the energy of the system was minimized by using the steepest descent algorithm. The system was then equilibrated in an isothermal-isochoric ensemble for 50 ns. The average temperature was maintained at 298 K using the velocity-rescale thermostat [34], with a time constant of 0.1 ps. This was followed by equilibration in the isothermal-isobaric ensemble for 50 ns, during which the average isotropic pressure was maintained at 1.0 bar using the Parrinello-Rahman [35] barostat, with a time constant of 2 ps. The Verlet list cutoff scheme, with a 10 Å cutoff, was used to treat Lennard-Jones interactions and short-range electrostatic interactions. The Particle-mesh Ewald (PME) [36] method with a grid spacing of 10 Å was used for long-range electrostatic interaction. The LINCS algorithm was used to constrain the bonds linked with hydrogen. MD simulation was performed using the Verlet leapfrog [37] integrator with a time step of 2 fs. The coordinates were saved every 10 ps. The total simulation time amounted to ~5.5 μs and ~6 μs for E1E2E3 and E1E3, respectively.

The simulation trajectories were preprocessed to remove solvent molecules and ions, and periodic boundary conditions were corrected. Further analyses were conducted using only the protein structure. The Root Mean Square Fluctuation (RMSF) was calculated with GROMACS, fitting each trajectory to the NPT-equilibrated structure. The S^2 order parameters were computed using the cpptraj [38, 39] module (version 6.18.1) from the AMBER software suite, and the backbone N-H bond vectors were used to determine the S^2 values after removing the global rotational and translational motion using GROMACS. The radius of gyration and backbone contact maps were computed using MDTraj [40] (version 1.9.7). Backbone contact maps were generated with a distance cutoff of 4.5 Å utilizing the Contact Map Explorer package. Hydrogen bonds were calculated using cpptraj with a distance cutoff of 3.0 Å and an angle cutoff of 135°. Salt bridge analysis is performed using VMD [41] (version 1.9.3) with a distance cutoff of 4.5 Å considering interactions between oxygen atoms of acidic residues and nitrogen atoms of basic residues. Cation-π interactions were identified with a distance cutoff of 6.0 Å between the "NZ" atom of lysine or the "CZ" atom of arginine and the centroid of the aromatic rings. π-π interactions were determined using a maximum heavy atom distance of 7.0 Å between aromatic rings, along with an angular criterion of ±36°, between the unit normal vectors of the rings, calculated using cpptraj.

To investigate the conformational space of the E1E2E3 and E1E3 protein, we have used a deep learning method known as autoencoder [42–45]. A mirror-symmetric architecture was utilized using tensorflow (version 2.9.1), backend with Keras python library (version 2.9.0), featuring 2048, 512, 256, 72, 36, and 12 neurons in the encoder and decoder's respective hidden layers with 2 neurons in the latent layer. This architecture was trained on MD simulation data of both proteins. As input features, we selected pairwise distances between Cα atoms, with every third residue skipped. The training dataset comprised 90% of the MD data, while the remaining 10% served as the validation set. The model was trained using a learning rate of 5e-4 with a batch size of 3000 for 300 epochs. The hyperbolic tangent (*tanh*) activation function was applied to all layers, with weights initialized using the Glorot uniform [46] method. The output layer, however, utilized a *sigmoid* activation function. The model was optimized using the Adam optimizer [47], with the mean squared error serving as the loss function. The resulting latent space was used to construct the free energy surface (FES). Gaussian Mixture Models (GMMs) were utilized to partition the free energy surface (FES) into distinct clusters. To determine the optimal number of clusters, the Bayesian Information Criterion (BIC) was applied. The BIC curve's slope was examined, and the "elbow" point, where the curve flattens, was identified as the optimal cluster count. This point represents a balance between achieving a good model fit and minimizing overfitting by avoiding model complexity.

2.6.1 | Quantification and Statistical Analysis

Data are expressed as means ± standard error of means (SEM). Statistical analyses were done using Graph Pad Prism (version 8.0). Student's *t*-test and ANOVA were used to determine statistical significance. A value of $p \leq 0.05$ was considered statistically significant. *$p \leq 0.05$; **$p \leq 0.05$; ***$p \leq 0.005$. In the figure, "*n*" refers to the number of animals per genotype included in each experiment ($n = 3$ means 3 WT and 3 KO animals), whereas "*N*" refers to the number of independent experimental cohorts performed.

3 | Results

3.1 | Temporal Regulation of Gluconeogenesis and Metabolic Flux During Starvation

The ability to mount an appropriate starvation response is critical not only for survival but also necessary to tune feed-fast cycles, which is inherently dependent upon metabolic plasticity and is largely dictated by hepatic physiology. Among others, hepatic gluconeogenesis is both essential for and diagnostic of anabolic-catabolic balance to sustain systemic metabolic needs. Despite decades of work that have identified upstream mechanisms that govern this, the temporal regulation of pyruvate utilization and its channeling into gluconeogenesis remains insufficiently understood. Toward this, we assessed gluconeogenic potential using the pyruvate tolerance test (PTT) as a function of duration of starvation in adult male mice, as indicated (Figure 1A,B and inset 1B). The PTT revealed significantly higher glucose production after 24 h of fasting compared to 6 h in response to the same dose of exogenously administered pyruvate. The inset illustrates the rise in blood glucose at 15 and 30 min relative to baseline (ΔTt–T0), reflecting the enhanced gluconeogenic capacity under prolonged fasting. Although anticipated and consistent with current literature [48–56], these results offer direct experimental evidence to posit dynamic regulation of metabolite channeling during progressive fasting states, which is less understood.

To gain further insights into differential metabolic channeling at 6- and 24-h of starvation (6hS and 24hS), we analyzed hepatic metabolome at baseline (PBS control) and in response to pyruvate administration (Figure 1C). As expected, the baseline metabolite profiles differed markedly between the 6hS and 24hS groups. Stable blood glucose (Figure 1B,E) and largely unchanged profiles of gluconeogenic intermediates hinted toward a predominant reliance on glycogenolysis at 6 h of starvation. However, pyruvate bolus at 6 h of starvation led to a significant increase in the levels of tricarboxylic acid (TCA) cycle intermediates, unlike the profile in the 24hS group that showed elevated TCA intermediates at baseline. In addition to indicating enhanced oxidative metabolism consistent with prolonged fasting, it was interesting to note that the 24-h baseline profile was comparable to the 6-h pyruvate administration. Moreover, following pyruvate administration, there was a marked reduction in key intermediates such as fructose-6-phosphate, glucose-6-phosphate, and Glycerol-3-phosphate, indicative of high pyruvate-dependent gluconeogenic potential (Figure 1D,E). These striking, pyruvate-dependent changes are significant across pathways, especially for glucose metabolism (Figure S1A–D). The differential re-wiring was further evident by changes in the ratios of key intermediates before and after pyruvate bolus (Figure 1F,G). For example, ratios of many TCA metabolites to gluconeogenic pathway intermediates such as αKG/G6P and OAA/G6P were distinct at 6 and 24 h of fasting. Importantly, these observations reveal biased substrate utilization, wherein gluconeogenic pathways are engaged at both 6 and 24 h of fasting. However, the metabolic fate of exogenously administered pyruvate differs markedly, reflecting starvation duration–dependent basal molecular programs that govern substrate utilization.

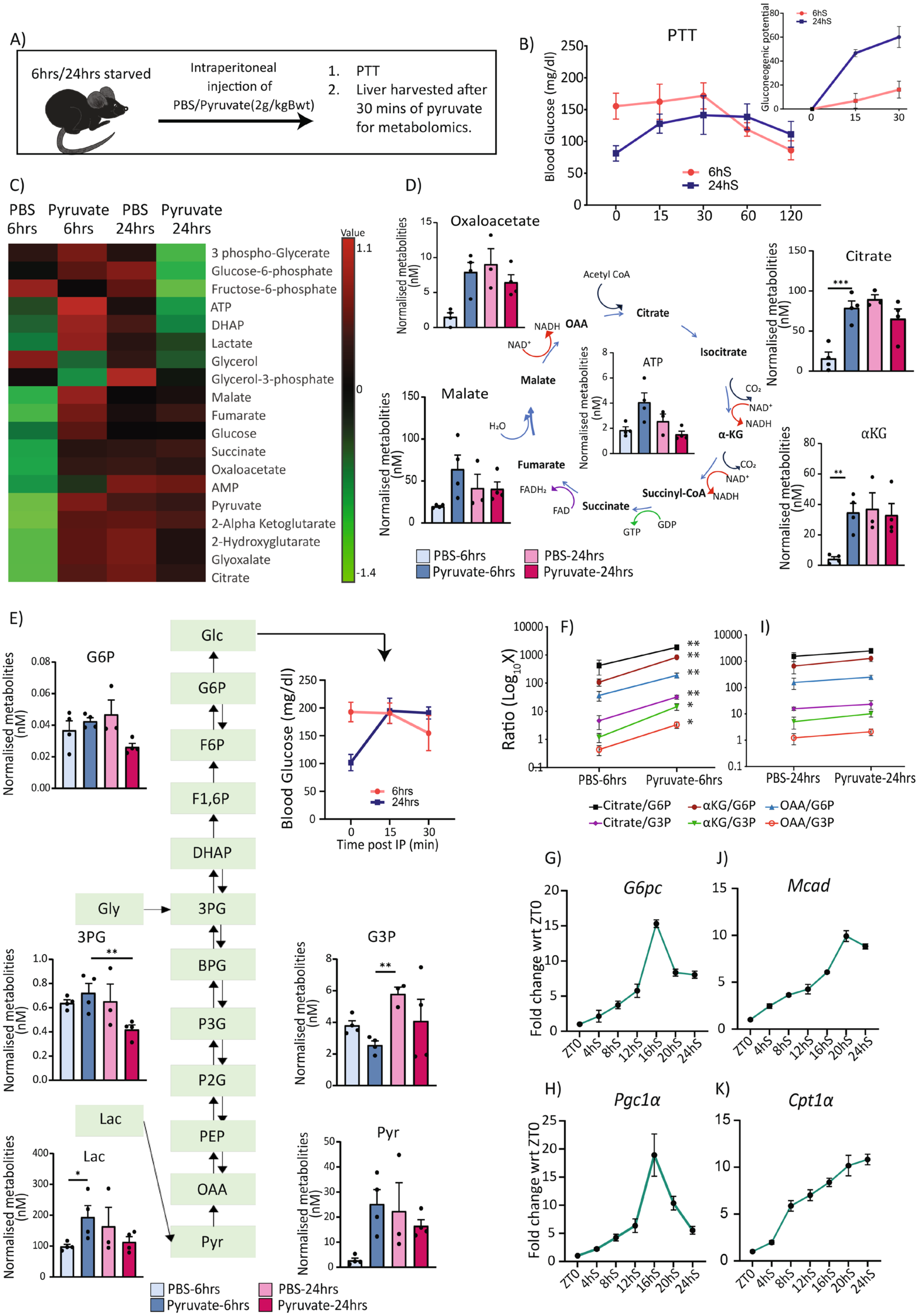

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FIGURE 1 | Gluconeogenesis and metabolic flux are temporally regulated during progressive starvation. (A) Schematic of the experimental design comparing 6- and 24-h starvation conditions. (B) The pyruvate tolerance test (2 g/kg body weight) was conducted at 6 and 24 h of starvation. The inset depicts the gluconeogenic potential, calculated as the increase in blood glucose at 15- and 30-min relative to the 0-min value for both fasting durations. The graph was plotted using GraphPad Prism 8.0. (C) Heatmap of normalized liver metabolite concentrations generated using MetaboAnalyst 5.0 ($n=4$). (D, E) Schematic diagrams of the gluconeogenic pathway and tricarboxylic acid (TCA) cycle annotated with normalized metabolite levels plotted on GraphPad Prism 8.0. (F, G) Ratios of metabolites, as indicated, were determined under PBS and pyruvate conditions at 6 and 24 h, respectively. Data are plotted on a Log_{10} scale, where "$Log_{10}X$" represents the log-transformed (base 10) ratio ($n=4$). (H–K) qPCR analysis of selected metabolic genes in livers from mice starved for the indicated durations. Transcript levels were normalized to β-actin ($N=2$, $n=3$); $n=3$–4 mice per genotype and $N=2$ independent cohorts.

3.2 | Dynamic Gene Expression Governs Progressive Starvation

Others and we have established the central role of transcriptional and post-transcriptional mechanisms in regulating starvation response during normal feed-fast cycles [49, 57–61]. However, the bi-directional interplay between hepatic metabolome and global gene expression cascades, and the cause-consequence mechanistic basis for precise temporal control of starvation response is much less understood. Toward this, we assessed for nodal genes whose expression is linked with glucose and fat metabolism across starvation phases. Gene expression analysis revealed a biphasic transcriptional response (Figure 1H–K and Sup. Figure S1E,F), with an early phase (0–12 h) marked by induction of gluconeogenic genes, followed by a later phase (12–24 h) enriched for fatty acid oxidation (FAO) genes, and importantly correlated well with the temporal change in hepatic metabolome (Figure 1C). Further, we assayed for levels and/or post-translational modifications of transcription factors at these time points, as indicated. We specifically found that phosphorylated CREB (pCREB) and SIRT1 peaked during the early starvation phase, while there was no change in the levels of FOXO1 and PPARα (Figure S1G,H). The temporal association between pCREB/SIRT1 induction and early gluconeogenic gene expression suggested that dynamic engagement of transcriptional regulators, rather than changes in their abundance alone, may contribute to the ordered progression of starvation-induced transcriptional programs.

3.3 | Temporal Control of SIRT1's Interactions by N-Terminal Disordered Region

Emerging literature has demonstrated the importance of intrinsically disordered regions, especially in transcription factors, in mediating transient protein–protein interactions and their deterministic role of IDRs in dictating promoter selectivity [5, 7–9]. Furthermore, it is necessary to note that there are no reports, that have shown the relevance of IDR-dependent dynamic switch in transcription factor engagement in an in vivo physiological setting, particularly during feed-fast cycles. Given this, we employed IUPred to compare the presence of IDRs in proteins, which are central for mounting a starvation response in the liver [62]. We found that while CREB and FOXO1 were predicted to contain multiple disordered regions, nuclear receptors such as PPARα and RXR appeared largely structured (Figure 2B,C and Figure S2A,D). However, it was interesting to note that co-regulators of transcription, viz. CRTC2, PPARGC1A (PGC-1α) and SIRT1 displayed a high degree and/or longer stretches of IDRs within them (Figure 2A–E and Figure S2A–E).

SIRT1 is known to act as a master regulator of transcription during starvation by modulating the activities of almost all the transcription factors and co-regulators [63–69]. In this context, it was tempting to investigate if the interactions of SIRT1 with these nodal factors are dynamically remodeled in response to starvation through co-immunoprecipitation from livers of mice starved for 3- and 12-h (3hS and 12hS). As shown in Figure 2F and consistent with earlier literature, we indeed observed a starvation-duration-dependent rewiring of the SIRT1 interaction. Notably, there was a distinguishable change wherein the interactions with CREB, PGC1α, and FOXO1 increased with prolonged fasting (Figure 2F). Given that the regulation of activity of key TFs converges at SIRT1, it led us to posit that competitive interactions may fine-tune SIRT1's TF selection and the downstream transcriptional output.

To test this, we employed luciferase reporter assays using minimal TF-responsive elements (PPRE, IRE, CRE). In concordance with the literature, SIRT1 activated PPRE and IRE elements but suppressed CRE-driven transcription (Figure S2F–H) [70–78]. Moreover, TF competition assays showed that SIRT1-mediated activation of PPRE (PPARα-dependent) and IRE (FOXO-1-dependent) is antagonized by FOXO1 and CREB, respectively (Figure 2G,H). This supported our hypothesis wherein SIRT1-TF selectivity drives competitive binding and hence, preferential promoter-specific transcription. This is also consistent with our earlier report that illustrated the pertinence of the N-terminal region of SIRT1 encoded by exon-2 in mediating interactions, which is differentially spliced out in a tissue-specific manner [25].

3.4 | Exon-2 Enhances N-Terminal Flexibility by Disrupting Inter-Domain Interactions

The results presented above prompted us to examine how exon-2 influences the conformational landscape of the SIRT1 N-terminus. To this end, we performed molecular dynamics simulations comparing the full-length N-terminal construct (E1E2E3, wild type) and the ΔE2 construct (E1E3, a previously identified splice variant) [25], as detailed in the Section 2. E1E2E3 displayed a higher radius of gyration ($\langle Rg \rangle = 32.6$ Å) than E1E3 ($\langle Rg \rangle = 28.1$ Å), consistent with a more extended conformation (Figure 3A). While partly due to the additional 39 residues of exon 2, analysis of secondary structure and contact

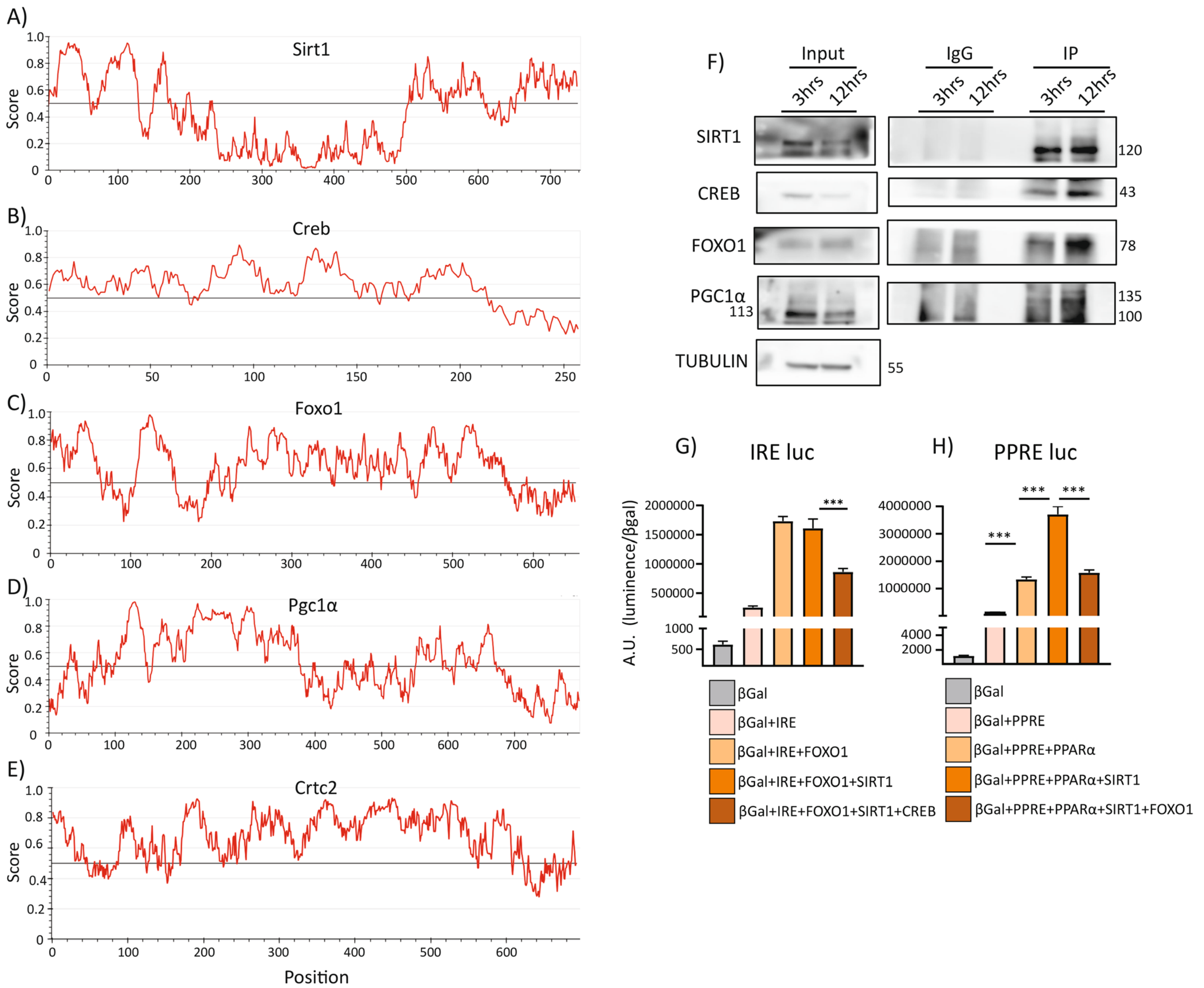

FIGURE 2 | N-terminal intrinsically disordered region in SIRT1 regulates its interactions with TFs and regulates starvation transcriptional response. (A–E) Intrinsic disorder prediction profiles of indicated proteins generated using IUPred. The horizontal threshold line at 0.5 distinguishes disordered (above) from structured (below) regions. (F) Co-immunoprecipitation using SIRT1 antibody from liver lysates collected after 3 and 12h starvation, probed for CREB, FOXO1, and PGC1α. (G, H) Reporter assays to test competitive binding effects on SIRT1-mediated activation or repression of target promoters ($n=3$, $N=3$). All graphs were plotted on GraphPad Prism 8.0; statistical analysis was done using two-way ANOVA; significance as $*p<0.05$; $**p<0.005$; $***p<0.0005$; $n=3$–4 mice per genotype and $N=3$ independent cohorts.

patterns became pertinent to reveal its impact on intra- and inter-domain interactions within the N-terminus. Strikingly, the presence of exon-2 influenced the structural properties of both exon-1 and exon-3, a hallmark of IDRs that are embedded within the structured proteins. Secondary structure analysis revealed that in E1E2E3, exon-1 (E1) showed decreased α-helical content (−12%) and increased coil (+13%), reflecting enhanced disorder, whereas exon-3 (E3) exhibited increased helicity (+5%) and reduced coils (−8%). Exon-2 (E2) itself remained predominantly coil-like (Figure 3A,B). These shifts paralleled contact map analyses: intra-domain contacts decreased within E1 (−9%) and increased within E3 (+13%) in E1E2E3, while E1–E3 inter-domain contacts were twofold higher in E1E3. Notably, E2 formed 3–6 fold more intra- than inter-domain contacts, suggesting that it acts as a flexible and largely independent module (Figure 3B and Figure S3C,D). Together, these data indicate that E2 promotes E1 disorder, stabilizes E3, and limits direct crosstalk between E1 and E3.

Dynamic properties reinforced this interpretation. Residue-wise RMSF showed higher flexibility in E1 when E2 was present (5–30Å in E1E2E3 vs. 5–20Å in E1E3), while E3 values remained comparable between the two structures (2–15Å) and E2 fluctuated within 3–15Å (Figure 3C,D). Consistently, S^2 order parameters for N-H bond vectors (0 = unrestricted motion, 1 = rigidity) revealed lower rigidity for E1 in E1E2E3 compared to E1E3, similar values for E3 with slightly higher C-terminal rigidity in E1E2E3, and high flexibility for E2 ($S^2<0.5$) was observed (Figure 3E–G).

These results were further substantiated by interaction analysis, which was assessed by hydrogen bonding. While

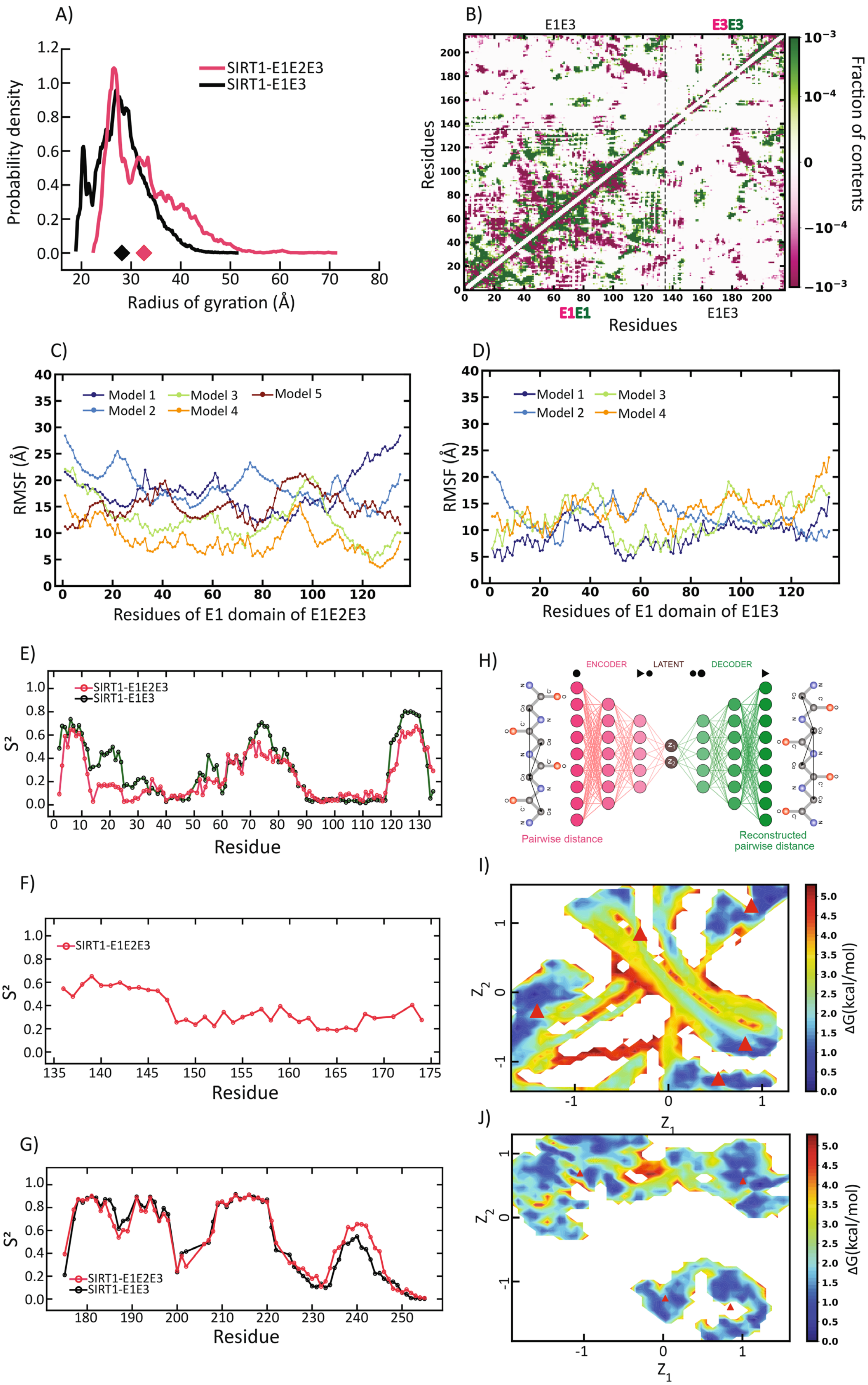

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FIGURE 3 | Structural Dynamics and free energy profiles of E1E2E3 and E1E3 proteins. (A) Distribution of Rg for the E1E2E3 and E1E3 protein. The diamond marker represents the average size of the two proteins. (B) Difference contact map of the E1 and E3 domain in E1E2E3 and E1E3. The higher the pink content, the greater the contacts in E1E3. The higher the green content, the greater the contacts in E1E2E3. RMSF of each residue for the E1 domain in (C) E1E2E3 (D) E1E3. N–H bond S^2 order parameter in E1E2E3 and E1E3 for (E) E1 domain (F) E2 domain (G) E3 domain. (H) Schematic of the autoencoder. Free energy surface of (I) E1E2E3 and (J) E1E3. The red triangles represent the GMM centers.

intra-domain hydrogen bonds in E1 and E3 were comparable between the two constructs, inter-domain E1–E3 bonds were 2.5–4 fold more abundant in E1E3 than in E1E2E3. E2 primarily formed intra-domain hydrogen bonds. Salt bridges, cation–π, and π–π interactions followed the same trend (Figures S3E–H and S4A).

Collectively, the reduction in E1–E3 contacts, increased coil propensity, elevated RMSF, and lower S^2 values demonstrate that exon-2 drives an extended and more disordered N-terminal ensemble. Free energy surface (FES) analysis further underscored this effect: Gaussian Mixture Model clustering identified four conformational states in E1E3 versus five in E1E2E3, each with distinct backbone contact maps (Figure 3H–J and Figure S4B,C). E2's intrinsic disorder modulated E1 compaction, such that flexible E2 promoted compact E1 states whereas rigid E2 increased disorder. Thus, exon-2 endows the SIRT1 N-terminus with structural plasticity by disrupting E1–E3 coupling and broadening the conformational landscape.

3.5 | SIRT1 Exon-2 Deletion Enhances CREB-Dependent Transcription During Fasting

Given that the exon-2 encoded part is disordered and since IDRs have been shown to be involved in dynamic protein–protein interactions, we wanted to assess the importance of SIRT1-IDR in this context. Toward this, we generated a lox-E2-lox line to obtain conditional loss of exon-2 in SIRT1, as described in the methods section. We specifically wanted to investigate the importance of SIRT1-IDR-dependent engagement with TFs during dynamic physiological transitions, viz., starvation. As shown in Figure S5A, we floxed out exon-2 in young adults using tamoxifen-induced Cre activation and loss of exon-2 was confirmed across tissues (Figure S1 and Figure 4A). Notably, mice expressing the endogenous SIRT1$^{\Delta E2}$ isoform remained viable, fertile, and displayed no overt developmental or behavioral abnormalities under basal conditions. Consistent with this, we did not observe significant differences in the weights of major metabolic tissues, including the liver, gonadal and subcutaneous white adipose depots, quadriceps, or gastrocnemius–soleus muscles, suggesting that loss of the exon 2-encoded region does not grossly compromise organismal health, adiposity, or muscle mass under basal conditions (Figure S5D).

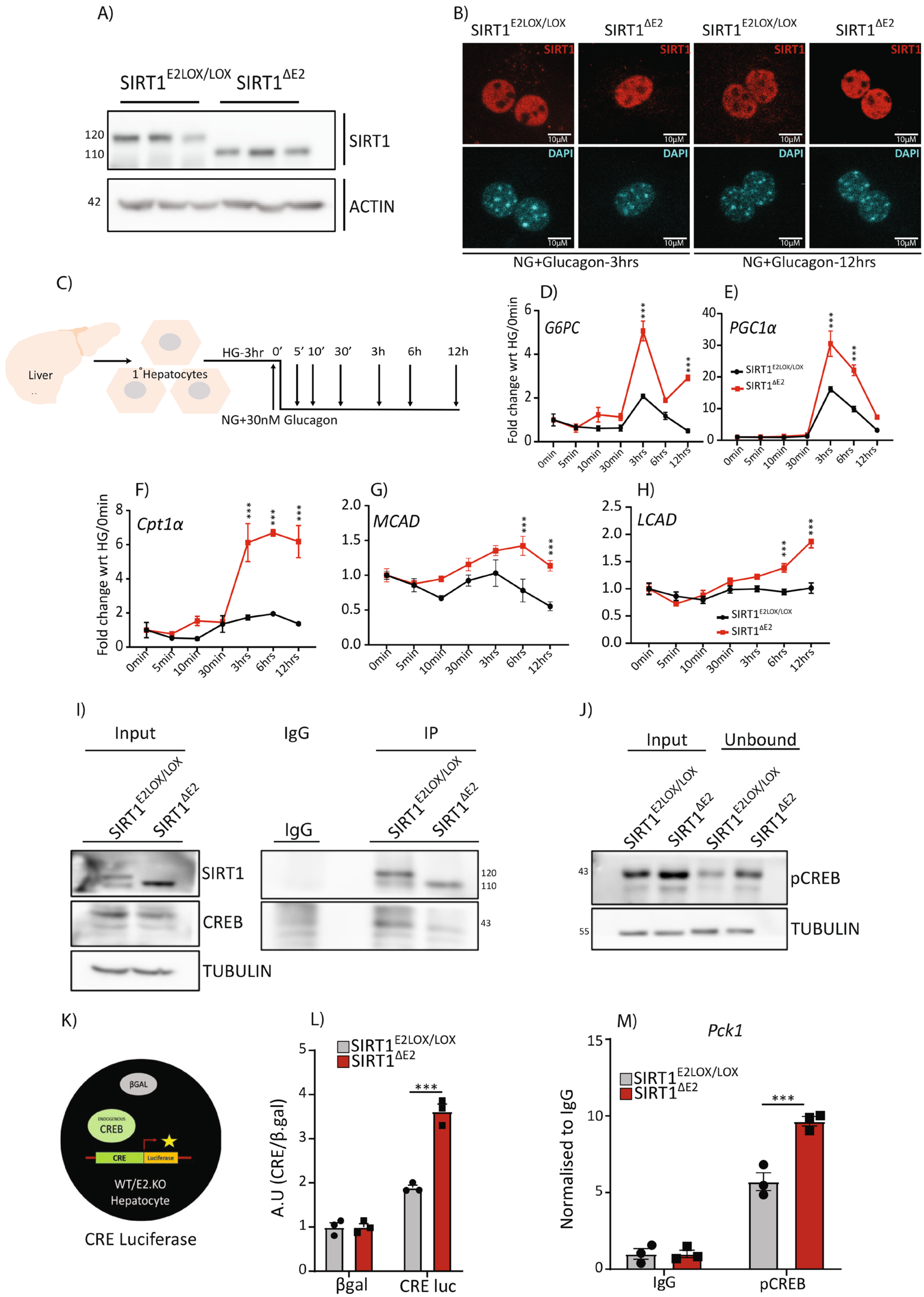
Others and we have shown that although SIRT1 is predominantly nuclear, it shuttles between cytosol and the nucleus, which is also determined by glycosylation within the exon-2, particularly in a fed state [58]. Hence, we assessed the localization of SIRT1$^{E2.lox/lox}$ and SIRT1$^{\Delta E2}$ under fasting-mimicking conditions (no-glucose media supplemented with 30 nM glucagon) and found that both SIRT1$^{E2.lox/lox}$ and SIRT1$^{\Delta E2}$ were predominantly localized in the nucleus (Figure 4B).

Our previous study identified SIRT1$^{\Delta E2}$ as a naturally occurring splice isoform lacking the exon 2-encoded region and extensively characterized its biochemical properties [25]. Using quantitative HPLC-based deacetylation assays with acetylated H3K9 peptide substrates, we demonstrated that SIRT1$^{\Delta E2}$ retained catalytic activity comparable to full-length SIRT1. Detailed kinetic analyses revealed similar substrate-dependent and NAD + -dependent enzymatic parameters, including Km and Vmax values. Importantly, these observations were reproduced using proteins purified from both bacterial and mammalian expression systems, indicating that the absence of exon 2 does not substantially alter the intrinsic deacetylase activity of SIRT1 [25]. These findings prompted us to investigate whether the exon 2-encoded intrinsically disordered region regulates fasting physiology at both molecular and physiological levels through modulation of molecular interactions rather than catalytic activity.

It is important to note that transcriptional induction of key gluconeogenic genes (*Pck1, G6pc1, and Pgc1α*) was significantly higher in SIRT1$^{\Delta E2}$ hepatocytes and clearly indicated that loss of IDR in SIRT1 N-terminus modulates the magnitude of the transcriptional response by enhancing glucagon-induced gene activation (Figure 4C–H).

This further led us to investigate the temporal dynamics of CREB activation and found rapid/heightened CREB phosphorylation at Ser133 in SIRT1$^{\Delta E2}$ hepatocytes in response to glucagon stimulation (Figure S5E). Moreover, this was consistent with the in vivo data from SIRT1$^{E2.lox/lox}$ and SIRT1$^{\Delta E2}$ mice following 12-h fasting (Figure S5F). Together, these data indicate that the loss of SIRT1 exon-2 leads to precocious and amplified CREB activation.

Even though CREB is necessary to elicit a fasting response, downstream of the glucagon-PKA axis, the mechanisms that govern temporal thresholding/restriction of CREB-dependent transcription are poorly understood. Based on the above results, we assayed for SIRT1-CREB interactions since SIRT1 is known to inhibit CREB and its cofactor CRTC2 [78, 79]. As shown in Figure 4I, co-immunoprecipitation assays revealed that SIRT1-bound CREB was dramatically lower in SIRT1$^{\Delta E2}$ compared to SIRT1$^{E2.lox/lox}$ in primary hepatocytes under fasting conditions. Exon-2 dependent interaction and consequent effect on CREB activity were further substantiated by increased levels of pCREB in the unbound fraction from SIRT1$^{\Delta E2}$ hepatocytes (Figure 4J).



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Further, this was supported by SIRT1 co-immunoprecipitation from the liver of progressively starved mice (3, 12, and 24 h). It was noted that across all time points, SIRT1$^{\Delta E2}$ mice exhibited consistently reduced CREB binding compared to SIRT1$^{E2.lox/lox}$ (Figure S6A–C), strengthening the finding that exon-2 is critical for maintaining SIRT1–CREB interaction during fasting.

FIGURE 4 | Exon-2 of SIRT1 is important for gatekeeping the starvation-induced gluconeogenic response by regulating the interaction between SIRT1 and CREB. (A) Western blots validation of SIRT1$^{\Delta E2}$ expression in the liver. (B) Representative immunofluorescence images confirming the nuclear localization of SIRT1$^{\Delta E2}$ under starvation conditions. (C) Schematic of the primary hepatocyte culture and glucagon treatment timeline. Cells were harvested at the indicated time points. (D–H) Time-course gene expression profiles of indicated glucagon-responsive genes in SIRT1$^{\Delta E2}$ and control hepatocytes ($n=3$, $N=3$). (I) Co-immunoprecipitation (Co-IP) of SIRT1 followed by immunoblotting for SIRT1 and CREB in SIRT1$^{E2.LOX/LOX}$ and SIRT1$^{\Delta E2}$ primary hepatocytes ($N=3$). (J) Input and unbound fractions from the Co-IP probed for pCREB ($N=3$). (K) Schematic of CREB-responsive element (CRE) luciferase reporter assay. (L) CRE reporter luciferase activity measured in SIRT1$^{E2.LOX/LOX}$ and SIRT1$^{\Delta E2}$ hepatocytes post-glucagon stimulation ($n=3$, $N=3$). (M) ChIP-qPCR for pCREB occupancy on *Pck1* promoter following glucagon treatment ($n=3$). All graphs were plotted on GraphPad Prism 8.0, one-way Anova was used for statistical testing (comparisons within the genotype only), significance as $^{*}p<0.05$; $^{**}p<0.005$; $^{***}p<0.0005$; $n=3$–4 mice per genotype and $N=3$ independent cohorts.

3.6 | SIRT1 Intrinsically Disordered Region Is Important for Gating CREB-Mediated Transcriptional Response

To determine the transcriptional consequences of this reduced interaction in a physiological setting, we performed CRE-luciferase reporter assays in SIRT1$^{E2.lox/lox}$ and SIRT1$^{\Delta E2}$ hepatocytes following glucagon stimulation. CRE-Luciferase activity was significantly elevated in SIRT1$^{\Delta E2}$ cells, confirming heightened CREB-mediated transcription in the absence of exon-2 in SIRT1 (Figure 4K,L).

To determine whether the effects of exon-2 deletion extended beyond CREB-dependent transcription, we next examined FOXO1- and PGC1α/PPARα-dependent transcriptional programs. As SIRT1-mediated regulation of these factors is known to depend on both deacetylation and physical interaction, we measured IRE- and PPRE-driven luciferase activity in primary hepatocytes. While CRE-luciferase activity was increased in SIRT1$^{\Delta E2}$ cells, both IRE- and PPRE-luciferase activities were significantly reduced compared to SIRT1$^{E2.lox/lox}$ controls (Figure S5G). These observations are consistent with our previous findings demonstrating reduced interaction of SIRT1$^{\Delta E2}$ with FOXO1 and PGC1α [25].

More importantly, chromatin immunoprecipitation (ChIP) for pCREB showed increased occupancy at the *Pck1* promoter in SIRT1$^{\Delta E2}$ hepatocytes (Figure 4M). This demonstrated that recruitment of active pCREB to the gluconeogenic gene promoters was regulated by SIRT1 exon-2.

In order to provide conclusive mechanistic evidence vis-à-vis SIRT1-IDR-mediated modulation of pCREB-dependent transcription, we pharmacologically inhibited CREB and PKA activity, as indicated. Pharmacological inhibition of PKA led to comparable reductions in gluconeogenic gene expression in both genotypes (Figure S7A,B), indicating that enhanced CREB activity is not associated with altered upstream PKA signaling in SIRT1$^{\Delta E2}$. However, it was striking to find that treatment of hepatocytes with the CREB inhibitor, 666–15, led to a significantly blunted induction of *Pck1, G6pc*, and *Pgc1α* in SIRT1$^{\Delta E2}$ cells, unlike in SIRT1$^{E2.lox/lox}$ hepatocytes (Figure 5A–C). This differential sensitivity to CREB inhibition clearly suggested that SIRT1$^{\Delta E2}$ hepatocytes exhibit heightened CREB activity akin to a "runaway" response.

The results described above prompted us to ascertain if this molecular rewiring led to a physiological output in terms of gluconeogenesis. As anticipated, SIRT1$^{\Delta E2}$ hepatocytes exhibited significantly elevated glucose production in response to both nutrient starvation and glucagon stimulation (Figure 5D,E and Figure S7C,D). These together illustrated that loss of SIRT1 exon-2 promotes a hyperactive gluconeogenic state, even under mild starvation, besides providing confirmatory data on the regulatory control exerted by SIRT1-IDR on regulating fasting response.

3.7 | SIRT1 IDR Mediated Gating Is Critical for Systemic Glycemic Control and Fed/Fast Cycles

Even though the results presented above clearly indicated the importance of SIRT1-IDR in programming gene expression during fasting, we wanted to investigate the in vivo physiological relevance of hepato-centric consequences of loss of exon-2 in SIRT1. As shown in Figure 5F, SIRT1$^{\Delta E2}$ mice exhibited increased body weight, compared to SIRT1$^{E2.lox/lox}$, in the absence of any measurable difference in feeding behavior (Figure 5G). Dynamic monitoring of blood glucose levels during fasting also revealed consistently elevated glucose in SIRT1$^{\Delta E2}$ mice (Figure 5H), even in the absence of exogenous nutrient intake. Next, we assessed systemic glucose homeostasis through glucose tolerance test (GTT), as described in the methods section. SIRT1$^{\Delta E2}$ mice displayed markedly impaired glucose clearance, with significantly increased area under the curve (AUC), and a failure to return to basal glycemic levels (Figure 5J, Figure S7E). Further, we employed the pyruvate tolerance test (PTT) and found exaggerated hyperglycemic response in SIRT1$^{\Delta E2}$ mice (Figure 5K and Figure S7F). Besides corroborating our in vitro and molecular data, these provide confirmatory evidence vis-à-vis the importance of SIRT1-IDR in orchestrating systemic metabolism.

3.8 | Liver Specific IDR Loss Disrupts Glucose Homeostasis/SIRT1 Exon-2 Deletion in Liver Drives Metabolic Imbalance

Others and we have earlier shown the importance of unraveling organ-system specific molecular mechanisms that drive organismal physiology, largely based on tissue specific knockouts or knockdowns. While IDRs have been shown to orchestrate transcription using in vitro cell culture models, there are no reports that have discovered the function of IDRs within specific tissues that impinge on whole body physiology. Hence, we generated liver-specific SIRT1 exon-2 knockout mice (SIRT1$^{hep\text{-}\Delta E2}$) by crossing SIRT1$^{E2.lox/lox}$ animals with Albumin-Cre mouse line. It was interesting to note

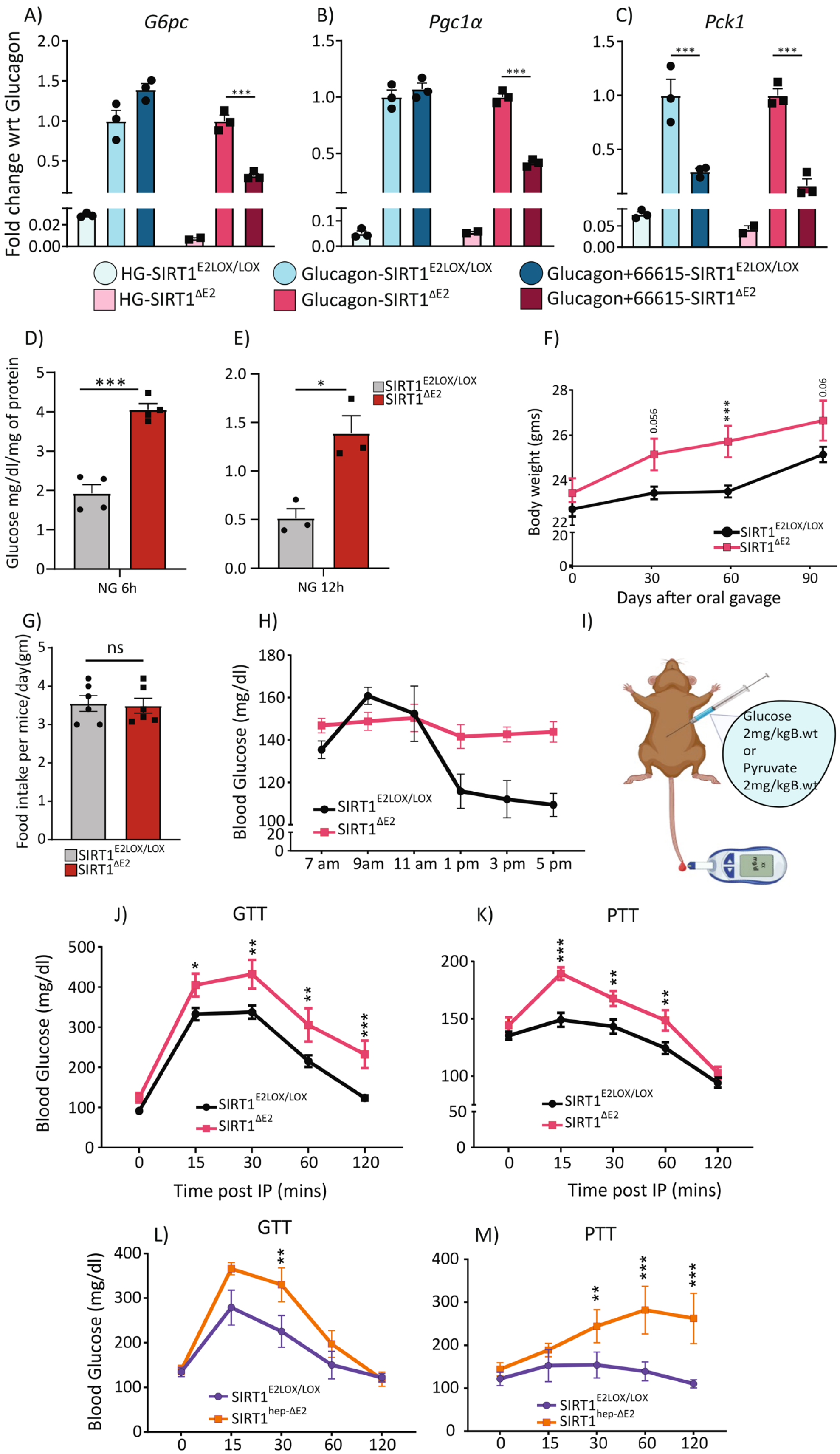

FIGURE 5 | Legend on next page.

FIGURE 5 | Loss of exon-2 in SIRT1 results in systemic glucose dyshomeostasis: (A–C) Relative expression of *G6pc, Pck1, and Pgc1α* after treatment with a CREB-specific inhibitor ($n=3$, $N=3$). (D–E) Gluconeogenesis assay was performed in $SIRT1^{E2LOX/LOX}$ and $SIRT1^{\Delta E2}$ hepatocytes cultured under NG (no glucose) (6h) and NG (12h) ($n=3$, $N=2$). (F) Body weights of $SIRT1^{\Delta E2}$ and $SIRT1^{E2.LOX/LOX}$ were measured under standard chow conditions ($n=4$–6). (G) Average daily food intake was quantified and compared between $SIRT1^{\Delta E2}$ and $SIRT1^{E2.LOX/LOX}$ ($n=3$, $N=2$). (H) Glucose measurements were taken every 2h from mice starting with food removal at ZT0 ($n=3$–5, $N=3$). (I) Schematic illustrating the Intraperitoneal injection protocol for glucose and pyruvate tolerance tests. (J, K) GTT and PTT were performed on 10–14-week-old male mice; $SIRT1^{\Delta E2}$ and $SIRT1^{E2.LOX/LOX}$ male mice following a 12-h overnight fast ($N=3$, $n=3$–5). (L, M) GTT and PTT were conducted in 10–14-week-old male $SIRT1^{hep\text{-}\Delta E2}$ and $SIRT1^{E2.LOX/LOX}$ mice, after a 12-h overnight fast ($n=17$). All graphs were plotted in GraphPad Prism 8.0, Student's *t*-test was used for statistical significance, $^{*}p<0.05$; $^{**}p<0.005$; $^{***}p<0.0005$; $n=3$–5/17 mice per genotype and $N=3$ independent cohorts.

that $SIRT1^{hep\text{-}\Delta E2}$ animals phenocopied the $SIRT1^{\Delta E2}$ mice vis-à-vis GTT and PTT (Figure 5L,M and Figure S7G,H). These exciting results indicated that functions of SIRT1-IDR in the liver contribute to the systemic metabolic dysregulation. More importantly, this illustrates tissue-autonomous modulation of SIRT1-TF interactions, mediated by the non-catalytic domain encoded by the exon-2, is necessary for starvation response at an organismal level. These results reveal, for the first time, that a non-catalytic IDR within SIRT1 mediates liver-specific transcriptional programs that are essential for maintaining whole-body metabolic balance. By demonstrating that loss of the SIRT1 IDR alters transcription factor engagement and downstream gene expression during fasting, our findings highlight an important role for intrinsically disordered regions within metabolic regulators. Together, these findings support the emerging view that disordered protein domains contribute to physiological adaptation through dynamic and context-dependent protein interaction networks.

4 | Discussion

Temporal engagement of transcription factors, which is critically dependent upon both cis-binding of upstream regulatory elements and trans interactions, is essential for orchestrating programmed reversible physiological transitions e.g., during fed-fast-refed cycles [2, 16, 80, 81]. Although the endocrine and metabolic signaling pathways that regulate TF activity are well established, the molecular mechanisms that coordinate promoter-selective TF recruitment during dynamic metabolic transitions remain incompletely understood [48, 51, 82–86]. Our findings reveal that loss of SIRT1 IDR disrupts metabolic fidelity, leading to premature transcriptional responses, aberrant glucose homeostasis, and impaired adaptation to fasting. Together, these results identify the SIRT1 IDR as a critical determinant of hepatic transcriptional responses to nutrient stress and provide insight into how dysregulated protein interactions contribute to metabolic dysfunction. More importantly, we provide hitherto unknown insights into the molecular origins of metabolic diseases mediated by protein disorder, which stem from dysregulated interactions.

The biphasic transcriptional response during starvation, characterized by early gluconeogenic gene induction followed by fatty acid oxidation, aligns with the dynamic rewiring of the SIRT1 interactome. Our findings suggest that competitive interactions between SIRT1 and distinct transcription factors contribute to the orderly progression of starvation-associated transcriptional programs. In this context, our work builds on previous studies, including that of Liu et al., which identified a temporal transition between CREB- and FOXO1-dependent transcriptional programs during fasting [78]. Our findings suggest that the SIRT1 IDR provides the conformational flexibility necessary to coordinate interactions with distinct transcriptional regulators across different phases of fasting. This is consistent with emerging paradigms in which IDRs facilitate multivalent, context-dependent interactions. While considerable attention has been given to the roles of IDRs in transcription factors [6, 15, 87], comparatively fewer studies have explored their function within transcriptional coactivators or metabolic regulators. Recent findings in p300/CBP suggest that IDRs in coactivators support multivalent interactions and enhance chromatin engagement [88–91], underscoring the broader relevance of such mechanisms in transcriptional regulation. Our work extends this paradigm to metabolic sensors, demonstrating that SIRT1's N-terminal IDR contributes to the regulation of transcriptional programs during nutrient stress. This aligns with reports showing that IDRs in nutrient-responsive enzymes such as ULK, RAPTOR, and CAMKII enable allosteric modulation, PTM-dependent regulation, and substrate selection. The observation that $SIRT1^{\Delta E2}$ hepatocytes exhibit unregulated CREB activity parallels findings in p53 [92–99], where disordered regions prevent promiscuous DNA binding by enforcing transient protein interactions. The resulting increase in gluconeogenic gene expression and systemic glucose dysregulation establishes a direct link between IDR-dependent interaction specificity and physiological output.

Importantly, liver-specific deletion of SIRT1's exon-2 recapitulated systemic glucose dysregulation, highlighting the tissue-autonomous role of this IDR in maintaining whole-body homeostasis. The failure of $SIRT1^{\Delta E2}$ mice to transition effectively during fasting states, marked by persistent gluconeogenic activation, suggests that the IDR acts as a conformational timer to buffer aberrant transcriptional cascades. This corroborates the broader evidence that IDRs in nutrient sensors like SIRT1 co-evolved with regulatory networks to integrate metabolic cues and transcriptional outputs.

The liver's ability to toggle between fed and fasted states hinges on remarkable transcriptional and metabolic plasticity, a process tightly regulated by nutrient sensors [60, 61, 100, 101]. Our findings reveal that the exon 2-encoded IDR of SIRT1 is indispensable for this dynamic adaptation, enabling a coordinated transcriptional response required to maintain hepatic metabolic homeostasis. Interestingly, although the $SIRT1^{\Delta E2}$ isoform is naturally expressed in several tissues, it is largely excluded from the liver [25]. In light of our findings, this tissue-specific splicing pattern may reflect the unique requirement of the liver for precise and rapidly adaptable transcriptional programs during fluctuating nutrient states. We therefore speculate that hepatic exclusion of the ΔE2 isoform preserves the interactional flexibility conferred by the SIRT1 IDR, thereby ensuring appropriate metabolic responses to fasting and refeeding. While the physiological significance of SIRT1$^{\Delta E2}$ expression in other tissues remains unclear, our findings suggest that tissue-specific expression of this isoform may diversify SIRT1 regulatory functions. This diversification could arise from altered protein–protein interaction networks in different cellular contexts.

These findings extend the study of IDRs beyond transcription factors to an upstream metabolic regulator, suggesting that structural disorder can tune interaction selectivity during physiological transitions. The exon 2-encoded IDR characterized here appears to be specific to SIRT1, and cannot be directly compared with analogous function in other sirtuins. However, the presence of predicted disordered segments in other family members, especially SIRT2 & SIRT6, raises the possibility that structural disorder may also contribute to functional diversification within the sirtuin family, which should be examined in the future [102–104]. In the context of highly conserved catalytic domains, such disordered regions may have evolved to expand the repertoire of interaction partners and substrates available to individual sirtuins, thereby enabling functional specialization across distinct physiological contexts.

Although our findings demonstrate that the SIRT1 exon 2-encoded IDR influences transcription factor-dependent responses during nutrient stress, the underlying interaction dynamics remain incompletely understood. Direct visualization of transcription factor recruitment and exchange in vivo will be required to establish the mechanistic basis of these effects. Additionally, while SIRT1$^{\Delta E2}$ mice exhibit clear metabolic abnormalities, more comprehensive analyses of body composition and energy expenditure will be valuable for defining the physiological consequences of exon 2 deletion, beyond hepatic metabolism.

Taken together, our study identifies a previously unappreciated role for the N-terminal disordered region in SIRT1 encoded by exon 2 in coordinating transcription factor activity during metabolic adaptation. Our findings demonstrate that disruption of this regulatory module perturbs temporal transcriptional responses despite largely preserved catalytic activity, highlighting an additional layer of control embedded within the structural organization of metabolic regulators. More broadly, these results support the emerging view that intrinsically disordered regions contribute to physiological regulation by shaping dynamic interaction networks and context-dependent signaling outputs. Future studies will be required to determine the extent to which similar mechanisms operate in other nutrient-sensing pathways and physiological transitions.

Author Contributions

Conceptualization: A.S. and U.K.-S. Methodology: A.S., S.A., J.M., C.B., S.C., A.K., and U.K.-S. Investigation: A.S., S.C. and A.K. Visualization: A.S., S.A., J.M., S.C. and C.B. Supervision: U.K.-S. Writing – original draft: A.S. and U.K.-S.

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Conflicts of Interest

Ullas Kolthur-Seetharam serves as an Editorial board member for the *FASEB* Journal and was recused from any editorial processes during the peer review of this manuscript, including the final decision on potential publication.

Data Availability Statement

All data needed to evaluate the conclusion of this paper are present in the paper and/or the Supporting Information.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Metabolic signatures of temporal starvation in mouse liver. **Figure S2:** Prediction of intrinsic disorder in key TFs and effect of SIRT1 on transcriptional activity of key starvation TFs. **Figure S3:** Structural and interaction analysis of E1E2E3 and E1E3 proteins. **Figure S4:** Inter-domain interactions and conformational states in E1E2E3 and E1E3 proteins. **Figure S5:** Molecular and Functional Validation of SIRT1$^{\Delta E2}$ and Its Impact on CREB Signaling. **Figure S6:** Dynamic Interaction of SIRT1 with CREB. **Figure S7:** Effect of PKA inhibitor on SIRT1$^{\Delta E2}$ and SIRT1$^{E2.LOX/LOX}$, functional consequences of loss of exon-2 on gluconeogenesis and area under the curve (AUC) of tolerance tests. **Table S1:** Key resource table.